# Take-Home Challenge

**Expected time: 3–4 hours • Free tools only • Present in 30 min**

## CONTEXT

For this challenge, we want to see how you think and build across the full stack of a small AI-powered product: data, logic, UI, and user experience. We're looking for product thinking, pragmatic technical choices, and the ability to ship something end-to-end.

## THE BRIEF

**Build a simple Account Intelligence app**

Imagine you're an Account Executive preparing for your day. You need to know: which accounts should I focus on, and what should I know before my next meeting? Build a small app that helps answer those two questions.

Your app should include the following layers:

1. **A data layer.** Use the sample dataset we provide (`account_data.csv` with ~1000 synthetic accounts — see the Sample Data section below for column details). You can enrich, transform, or extend this data however you see fit.
2. **A scoring or intelligence layer.** Build some logic that helps prioritize which accounts need attention. We care more about your reasoning for the approach than the sophistication of the model.
3. **An AI-generated insight.** For at least one account, use an LLM (any free API: OpenAI free tier, Anthropic free tier, Groq, Ollama locally, or any other) to generate something useful: a meeting prep brief, a risk summary, suggested talking points, or a short account narrative. Show us how you think about prompting and what context you feed the model.
4. **A simple UI.** A frontend that displays the prioritized accounts and the AI-generated insight. We're looking for usability, not design. Use Lovable, Cursor (free trial), Streamlit, Gradio, or any free tool.

## SAMPLE DATA

We provide `account_data.csv` with ~1000 synthetic accounts. The data is designed to resemble a real-world scenario — treat it as you would any production dataset.

| Column | Type | Description |
|---|---|---|
| account_id | string | Unique identifier (ACC-0001 to ACC-1000) |
| account_name | string | Company name |
| account_description | string | One-sentence description of what the company does |
| industry | string | Industry vertical (Technology, Healthcare, Finance, Retail, Manufacturing, Education, Media) |
| segment | string | Customer segment (Enterprise, Mid-Market, SMB) |
| region | string | Geographic region (NA, EMEA, APAC, LATAM) |
| nr_employees | integer | Number of employees at the company |
| days_to_next_renewal | integer | Days until contract renewal |
| days_since_last_sales_activity | integer | Days since last sales touchpoint |
| nr_support_tickets | integer | Open support tickets |
| ai_usage | float | AI feature adoption score (0.0–1.0) |
| nr_active_users | integer | Number of active users on the platform |
| nr_licensed_seats | integer | Number of licensed seats |
| current_revenue | float | Current annual revenue from this account ($) |
| call_transcript_summary | string | Summary of most recent sales call |

| Column | Type | Description |
|---|---|---|
| revenue_end_of_quarter | float | Actual revenue at end of quarter ($) |

## TOOLS

Use only free tools. Here are some suggestions, but you're free to use anything that's free and accessible:

- **UI**: Streamlit, Gradio, Lovable, Retool (free tier)
- **LLM**: OpenAI free tier, Anthropic free tier, Groq, Google Gemini, Ollama (local)
- **IDE / coding assistants**: Cursor (free trial), VS Code + Copilot, Windsurf, or any IDE you prefer

## WHAT WE'RE EVALUATING

We're not grading your code quality or your model accuracy. We're evaluating how you think about building a product. Specifically:

- **Product thinking**: Did you build something an AE would actually find useful?
- **Scoring rationale**: Can you explain *why* your prioritization logic makes sense, not just *how* it works?
- **Prompt design**: What context did you give the LLM, and why? How did you structure the prompt to get useful output?
- **End-to-end delivery**: Does the app work? Can we run it from your README and see results?
- **Communication**: Is your README clear? Can someone unfamiliar with your code understand your approach?
- **Pragmatic trade-offs**: Did you scope wisely? Can you articulate what you'd do differently with more time?

## DELIVERABLES

- Submit a public **GitHub repository**
- Your repo must include a **README** covering:
    - What the app does
    - How to set it up and run it
    - A brief architecture overview (diagram optional but welcome)
    - Key decisions you made and why
- Your code should be **runnable** by following the README instructions.

## PRESENTATION (30 MINUTES)

You'll present your work to 2–3 members of our team. Here's the format:

**Demo (10 min)**: Do a **live demo**. Walk us through the app as if we're AEs seeing it for the first time. What would we see? What would we do with it? Slides are optional and limited to 5 max.

**Decisions (10 min)**: Walk us through your choices. How did you build the scoring logic and why? What context did you give the LLM and why?

**Discussion (10 min)**: We'll ask questions. These will be product, design and technical questions.

## IMPORTANT NOTES

- **Don't over-engineer.** We value clear reasoning and pragmatic choices over polish or sophistication.
- **You can use AI assistants** (Copilot, ChatGPT, Claude, etc.) — just be ready to explain every line of your code.
- **No need to deploy.** A local demo that runs from your README instructions is perfectly fine.

## TIMELINE

Please submit your work within 5 calendar days. We give you 5 days so you can fit the 3–4 hours into your schedule — not so you spend all 5 days on it. Your presentation will be scheduled within a few days of submission.

We're excited to see how you think. Build something small, explain it well, and show us you can own a problem end-to-end. Good luck